

SIMPLE APPROACH

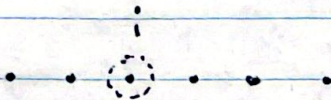
using a 1D Ising model Hamiltonian:

$$H(\sigma) = -J \sum_{\langle ij \rangle \text{ nearest neighbor}} \sigma_i \sigma_j$$

define syndromes:

$$\sigma_1 \quad \sigma_2 \quad \sigma_3 \quad \sigma_4 \quad \sigma_5 \quad \sigma_6 \quad \dots$$

$$S_i = \sigma_i \oplus \sigma_{i+1} \quad \text{w/ } \sigma_i \in \{0, 1\}$$



hamiltonian term involving site i will be $-J[\sigma_{i-1}\sigma_i + \sigma_i\sigma_{i+1}]$ (*)
in terms of syndromes:

well if we let $\sigma_i \in \{1, -1\}$ instead we could simplify

$$S_i = \sigma_i \sigma_{i+1} \quad \text{and if } S_i = 1 \text{ aligned, } S_i = -1 \text{ anti-aligned}$$

then (*) becomes $-J[S_{i-1} + S_i]$

ΔH_i → change in energy if a bit is flipped at site S_i

$$\Delta H_i = H'_i - H_i = -J[S'_{i-1} + S'_i] + J[S_{i-1} + S_i]$$

$$\text{and } \left. \begin{array}{l} S'_{i-1} = -1 \times S_{i-1} \\ S'_i = -1 \times S_i \end{array} \right\} = 2J[S_{i-1} + S_i]$$

if $\Delta H_i < 0$ energy decreased by flip → accept change

$\Delta H_i > 0$ energy increase by flip → reject change

$\Delta H_i = 0$ accept energy w/ 50% prob.

Qs: • how to apply this? on random spins or sequentially?

• Boundary

$$\left. \begin{array}{l} \Delta H_0 = -J[S'_0] + JS_0 = 2JS_0 \\ \Delta H_N = 2JS_N \end{array} \right\} \begin{array}{l} \text{row 0 spin} \\ \text{is fixed} \end{array}$$